



Computational Thinking

A. BHULAI



Project assignment

- Report & presentation
- Deadline: 13 Dec. 11.00 PM
- Working on project: 4, 5, 11 December (Quiz 2 is also on 5 Dec.!)
- Project presentation: 12 December (announcement will follow)
- Check also the grading scheme in Canvas
- TAs are available for questions

Minimum requirements report

- State the names of the group members and the group number.
- What is the problem you are solving?
- How have you tackled the problem (which algorithms, solution strategies, tools, etc.)?
- Which problems have you encountered in solving the problem?
- How have you solved those problems (which solution strategies/algorithms)?
- How did you split the tasks?
- Specify who did what.
- How much time did you spend on solving the problem (an estimation is sufficient)?

Minimum requirements presentation

- The presentation should not be longer than 10 min.
- State who the group members are and who did what.
- State the problem that you are going to solve.
- Show which solution strategies or algorithms you have used to tackle the problem.
- Show the steps!!!
- Discuss any problems that you have encountered in solving the problem.
- Discuss any solutions to the problems you have encountered.
- Show the solution (if you have found one).
- All group members should be present.

Project *iTrade*

Task *iTrade*

- Recommend a given number of stocks for user to invest in.
- Recommend the top n highest performing stocks to the user.
- Output a ranked list of the stocks based on performance.

Three preference settings

1. Highest in the industry
2. Establishment year (1800-2020)
3. ESG (environment, social, governance) criteria

Project *iTrade*: Your task

1. Think about the broader implications of modern computing technology on the financial world and briefly express your thoughts on the topic in the *Context Task* described in the assignment.
2. Design an algorithm that represents your implementation of *iTrade*.
3. Provide a flowchart for this algorithm.
4. Provide pseudocode (the steps) for this algorithm.
5. Implement your algorithm in a Python program.

See for details “Project assignment: *iTrade*” on Canvas.

Procedure

- ❑ We will discuss the problem
- ❑ You work together in pairs
- ❑ 5 -10 minutes per problem
- ❑ We discuss the solution

Problem: Online patient database

A general practice in Amsterdam moves from a paper-based patient record to an online database. The practice has data on 2374 patients. In the online database, the records need to be sorted alphabetically based on the name of the patient.

For which sorting algorithm is the *difference* in computing time largest between the worst-case and the best-case scenario for this dataset: bubble sort, merge sort, or quicksort?

Solution: Online patient database

Algorithm	Worst-case	Best-case	Difference
Bubble sort	$n^2 = 5\,635\,876$	$n = 2\,374$	5 633 502
Merge sort	$n * \log_2 n \approx 26\,620$	$n * \log_2 n \approx 26\,620$	0
Quicksort	$n^2 = 5\,635\,876$	$n * \log_2 n \approx 26\,620$	5 609 256

Problem: Medical electronic system

A hospital introduces a new electronic system in which nurses can log different types of procedures and medications which they administer to the patients. For each patient, they need to enter the name of the procedure/medication immediately after administering it. However, to help quicker lookup, the daily data for each patient is stored in an alphabetical order, not by administration time. At the end of a given day, one patient has the following log:

- ❖ Measure_body_temperature
- ❖ Sumamed
- ❖ Vitamin_C
- ❖ Ambrobene
- ❖ Kalmopyrin

The electronic log sorts this list into alphabetical order using quicksort. Show the intermediate steps of the algorithm on this list.

Solution: Medical electronic system

Abbreviate the log: M, S, V, A, K

- Pivot: M
The result of the first iteration: K, A, M, V, S
- Resulting sub-problems:
 - Sorting K, A. Pivot is K → results in sorted list A, K.
 - Sorting V, S. Pivot is V → results in sorted list S, V.
- Combining the sorted lists: A, K + M + S, V → the sorted list is A, K, M, S, V.
- Answer: the electronic log stores the entries in the following order:
 1. Ambrobene
 2. Kalmopyrin
 3. Measure_body_temperature
 4. Sumamed
 5. Vitamin_C

Problem:

A hospital introduces a new electronic system in which nurses can log different types of procedures and medications which they administer to the patients. For each patient, they need to enter the name of the procedure/medication immediately after administering it. However, to help quicker lookup, the daily data for each patient is stored in an alphabetical order, not by administration time. At the end of a given day, one patient has the following log:

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